

O-RAN VIP Suite — Detailed Test Plan (vplan) & Coverage

100% PASS

Author: AVIK MAJUMDAR · AVIK VIP FACTORY · 13 components (4G CPRI-over-Eth + 5G NR eCPRI Split 7.2x) + uvm_reg RAL · generated 2026-08-11 04:10 UTC
Methodology: constrained-random / coverage-driven. Every test class executed on Verilator 5.050 (+5.020); STATUS from simulator evidence only.

1523/1523

executed tests, 100% pass

100%

functional coverage (merged)

100% / 100%

code stnt / DUT branch (eff)

13 + RAL

components verified

1 · Verification strategy

Each component is modelled at the 64-bit protocol-header transaction level with a single `spec.json` descriptor (field widths, legal space, cross-field constraints, coverage bins) driving all lanes. Stimulus is **constrained-random**: the legal space and corner weighting come from the protocol spec, never from hardcoded values; the same seed reproduces a run exactly. Coverage is **coverage-driven** — coverpoints trace to vplan features, holes are closed by directed constraints then reverted. Five executed test classes per component:

Class	Purpose	Stimulus	Gate	Pass definition
smoke	bring-up sanity	seed-locked constrained-random (1/2/3)	GATE 3	legal txn, UVM_ERROR=0
random	constrained-random soak	100 seeds, legal-space + cross constraints	GATE 5	0 legality/cross violations
negative	error injection	constrained illegal space (inject_illegal)	GATE 6	violation DETECTED (EXPECTED_FAILURE_DETECTED)
directed	coverage-hole / FR targeting	inline constraint on the hole	GATE 4/8	targeted bin hit
corner	boundary / wrap	boundary-weighted dist{}	GATE 8	all boundary bins hit, legal

2 · Traceability model

FR-### (requirement) → **FEAT-###** (feature) → **VC/TC-###** (test case & class) → **SEQ-###** (constrained-random sequence) → **COV-###** / **CHK-###** / **SVA** (coverage bin & checker) → **BUG-###** (if a failure is found).

Each per-component table below realises this chain: every field yields a FEAT (legal-space + legality checker + coverpoint), every cross-constraint yields a FEAT + cross-coverage + SVA invariant, plus pack/round-trip and reset features.

Jump to component: ecpri_transport cpri_eth uplane cplane splane mplane beamforming compression prach mimo_massive bwp mmwave laa ral

3 · Per-component test plan

ecpri_transport — eCPRI Transport Layer (4G+5G)

Spec: eCPRI v2.0 §3.1 common header + §3.2 Type-0 IQ · fields: 9 · cross-constraints: 0 · coverpoints: 5 · result: **40/40 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-ECPRI_TRANSPORT-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-ECPRI_TRANSPORT-01
FEAT-ECPRI_TRANSPORT-02	Field <code>rsvd</code> [3b] legal-space coverage & legality check	FR-ECPRI_TRANSPORT-02
FEAT-ECPRI_TRANSPORT-03	Field <code>concat</code> [1b] legal-space coverage & legality check	FR-ECPRI_TRANSPORT-03
FEAT-ECPRI_TRANSPORT-04	Field <code>msg_type</code> [8b] legal-space coverage & legality check	FR-ECPRI_TRANSPORT-04
FEAT-ECPRI_TRANSPORT-05	Field <code>payload_size</code> [16b] legal-space coverage & legality check	FR-ECPRI_TRANSPORT-05
FEAT-ECPRI_TRANSPORT-06	Field <code>pc_id</code> [16b] legal-space coverage & legality check	FR-ECPRI_TRANSPORT-06
FEAT-ECPRI_TRANSPORT-07	Field <code>seq_id</code> [8b] legal-space coverage & legality check	FR-ECPRI_TRANSPORT-07
FEAT-ECPRI_TRANSPORT-08	Field <code>e_bit</code> [1b] legal-space coverage & legality check	FR-ECPRI_TRANSPORT-08
FEAT-ECPRI_TRANSPORT-09	Field <code>sub_seq</code> [7b] legal-space coverage & legality check	FR-ECPRI_TRANSPORT-09
FEAT-ECPRI_TRANSPORT-10	64-bit header pack/unpack round-trip integrity	FR-ECPRI_TRANSPORT-10
FEAT-ECPRI_TRANSPORT-11	Reset / power-on default values	FR-ECPRI_TRANSPORT-11

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-ECPRI_TRANSPORT	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-ECPRI_TRANSPORT	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	20/20	PASS
TC-NEG-ECPRI_TRANSPORT	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-ECPRI_TRANSPORT	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	14/14	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-003	<code>concat</code>	bool	bins {0}, {1} (2 bins)	100%
COV-004	<code>msg_type</code>	enum	one bin per value: {0, 1, 2, 3, 4, 5, 6, 7} (8 bins)	100%
COV-005	<code>payload_size</code>	bucket4	4 range buckets over [8:1024] (~254/bucket)	100%
COV-007	<code>seq_id</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%
COV-008	<code>e_bit</code>	bool	bins {0}, {1} (2 bins)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-ECPRI_TRANSPORT-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-ECPRI_TRANSPORT-02	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-ECPRI_TRANSPORT-03	<code>concat</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-ECPRI_TRANSPORT-04	<code>msg_type</code> legality: value [0 : 7]	scoreboard + SVA	exercised
CHK-ECPRI_TRANSPORT-05	<code>payload_size</code> legality: value [8 : 1024]	scoreboard + SVA	exercised
CHK-ECPRI_TRANSPORT-06	<code>pc_id</code> legality: value [0 : 65535]	scoreboard + SVA	exercised
CHK-ECPRI_TRANSPORT-07	<code>seq_id</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-ECPRI_TRANSPORT-08	<code>e_bit</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-ECPRI_TRANSPORT-09	<code>sub_seq</code> legality: value [0 : 127]	scoreboard + SVA	exercised
CHK-ECPRI_TRANSPORT-RT	pack→unpack round-trip == original	scoreboard	exercised

cpri_eth — CPRI-over-Ethernet (4G LTE)

Spec: CPRI-over-Ethernet 4G (IEEE1914.3 RoE + eCPRI transport) · fields: 10 · cross-constraints: 0 · coverpoints: 5 · result: **122/122 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-CPRI_ETH-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-CPRI_ETH-01
FEAT-CPRI_ETH-02	Field <code>bw_profile</code> [4b] legal-space coverage & legality check	FR-CPRI_ETH-02
FEAT-CPRI_ETH-03	Field <code>direction</code> [1b] legal-space coverage & legality check	FR-CPRI_ETH-03
FEAT-CPRI_ETH-04	Field <code>iq_rate_id</code> [4b] legal-space coverage & legality check	FR-CPRI_ETH-04
FEAT-CPRI_ETH-05	Field <code>synce_ql</code> [4b] legal-space coverage & legality check	FR-CPRI_ETH-05
FEAT-CPRI_ETH-06	Field <code>frame_id</code> [10b] legal-space coverage & legality check	FR-CPRI_ETH-06
FEAT-CPRI_ETH-07	Field <code>subframe</code> [4b] legal-space coverage & legality check	FR-CPRI_ETH-07
FEAT-CPRI_ETH-08	Field <code>cm_flag</code> [1b] legal-space coverage & legality check	FR-CPRI_ETH-08
FEAT-CPRI_ETH-09	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-CPRI_ETH-09
FEAT-CPRI_ETH-10	Field <code>rsvd</code> [24b] legal-space coverage & legality check	FR-CPRI_ETH-10
FEAT-CPRI_ETH-11	64-bit header pack/unpack round-trip integrity	FR-CPRI_ETH-11
FEAT-CPRI_ETH-12	Reset / power-on default values	FR-CPRI_ETH-12

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-CPRI_ETH	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-CPRI_ETH	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-CPRI_ETH	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-CPRI_ETH	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	16/16	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-002	<code>bw_profile</code>	enum	one bin per value: {0, 1, 2, 3, 4, 5} (6 bins)	100%
COV-003	<code>direction</code>	bool	bins {0}, {1} (2 bins)	100%
COV-005	<code>synce_ql</code>	enum	one bin per value: {0, 2, 4, 11, 15} (5 bins)	100%
COV-008	<code>cm_flag</code>	bool	bins {0}, {1} (2 bins)	100%
COV-009	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-CPRI_ETH-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-CPRI_ETH-02	<code>bw_profile</code> legality: value ∈ {0, 1, 2, 3, 4, 5}	scoreboard + SVA	exercised
CHK-CPRI_ETH-03	<code>direction</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-CPRI_ETH-04	<code>iq_rate_id</code> legality: value [0 : 15]	scoreboard + SVA	exercised
CHK-CPRI_ETH-05	<code>synce_ql</code> legality: value ∈ {0, 2, 4, 11, 15}	scoreboard + SVA	exercised
CHK-CPRI_ETH-06	<code>frame_id</code> legality: value [0 : 1023]	scoreboard + SVA	exercised
CHK-CPRI_ETH-07	<code>subframe</code> legality: value [0 : 9]	scoreboard + SVA	exercised
CHK-CPRI_ETH-08	<code>cm_flag</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-CPRI_ETH-09	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-CPRI_ETH-10	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-CPRI_ETH-RT	pack→unpack round-trip == original	scoreboard	exercised

uplane — U-Plane Split 7.2x (5G NR)

Spec: O-RAN.WG4.CUS §7 U-plane Section Types 1/3/5/6 · fields: 10 · cross-constraints: 1 · coverpoints: 5 · result: **123/123 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-UPLANE-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-UPLANE-01
FEAT-UPLANE-02	Field <code>section_type</code> [4b] legal-space coverage & legality check	FR-UPLANE-02
FEAT-UPLANE-03	Field <code>numerology</code> [4b] legal-space coverage & legality check	FR-UPLANE-03
FEAT-UPLANE-04	Field <code>start_prb</code> [9b] legal-space coverage & legality check	FR-UPLANE-04
FEAT-UPLANE-05	Field <code>num_prb</code> [9b] legal-space coverage & legality check	FR-UPLANE-05
FEAT-UPLANE-06	Field <code>symbol_id</code> [4b] legal-space coverage & legality check	FR-UPLANE-06
FEAT-UPLANE-07	Field <code>comp_type</code> [2b] legal-space coverage & legality check	FR-UPLANE-07
FEAT-UPLANE-08	Field <code>comp_bits</code> [5b] legal-space coverage & legality check	FR-UPLANE-08
FEAT-UPLANE-09	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-UPLANE-09
FEAT-UPLANE-10	Field <code>rsvd</code> [15b] legal-space coverage & legality check	FR-UPLANE-10
FEAT-UPLANE-11	Cross-field constraint: <code>start_prb + num_prb <= 275</code>	FR-UPLANE-11
FEAT-UPLANE-12	64-bit header pack/unpack round-trip integrity	FR-UPLANE-12
FEAT-UPLANE-13	Reset / power-on default values	FR-UPLANE-13

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-UPLANE	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-UPLANE	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-UPLANE	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-UPLANE	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	17/17	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-002	<code>section_type</code>	enum	one bin per value: {1, 3, 5, 6} (4 bins)	100%
COV-003	<code>numerology</code>	enum	one bin per value: {0, 1, 2, 3, 4} (5 bins)	100%
COV-005	<code>num_prb</code>	bucket4	4 range buckets over [1:275] (~68/bucket)	100%
COV-007	<code>comp_type</code>	enum	one bin per value: {0, 1, 2, 3} (4 bins)	100%
COV-009	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%
COVX-UPLANE-01	<code>cross</code>	cross	legal × illegal region of (<code>start_prb + num_prb <= 275</code>)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-UPLANE-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-UPLANE-02	<code>section_type</code> legality: value ∈ {1, 3, 5, 6}	scoreboard + SVA	exercised
CHK-UPLANE-03	<code>numerology</code> legality: value [0 : 4]	scoreboard + SVA	exercised
CHK-UPLANE-04	<code>start_prb</code> legality: value [0 : 273]	scoreboard + SVA	exercised
CHK-UPLANE-05	<code>num_prb</code> legality: value [1 : 275]	scoreboard + SVA	exercised
CHK-UPLANE-06	<code>symbol_id</code> legality: value [0 : 13]	scoreboard + SVA	exercised
CHK-UPLANE-07	<code>comp_type</code> legality: value ∈ {0, 1, 2, 3}	scoreboard + SVA	exercised
CHK-UPLANE-08	<code>comp_bits</code> legality: value [9 : 16]	scoreboard + SVA	exercised
CHK-UPLANE-09	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-UPLANE-10	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-UPLANE-11	cross invariant: <code>start_prb + num_prb <= 275</code>	scoreboard + SVA	exercised
CHK-UPLANE-RT	pack→unpack round-trip == original	scoreboard	exercised

cplane — C-Plane Split 7.2x (5G NR)

Spec: O-RAN.WG4.CUS §5 C-plane Section Types 0-8, Ext 1-11 · fields: 10 · cross-constraints: 1 · coverpoints: 6 · result: **123/123 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-CPLANE-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-CPLANE-01
FEAT-CPLANE-02	Field <code>section_type</code> [4b] legal-space coverage & legality check	FR-CPLANE-02
FEAT-CPLANE-03	Field <code>section_ext</code> [4b] legal-space coverage & legality check	FR-CPLANE-03
FEAT-CPLANE-04	Field <code>start_symbol</code> [4b] legal-space coverage & legality check	FR-CPLANE-04
FEAT-CPLANE-05	Field <code>num_symbol</code> [4b] legal-space coverage & legality check	FR-CPLANE-05
FEAT-CPLANE-06	Field <code>num_sections</code> [4b] legal-space coverage & legality check	FR-CPLANE-06
FEAT-CPLANE-07	Field <code>beam_id</code> [16b] legal-space coverage & legality check	FR-CPLANE-07
FEAT-CPLANE-08	Field <code>numerology</code> [4b] legal-space coverage & legality check	FR-CPLANE-08
FEAT-CPLANE-09	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-CPLANE-09
FEAT-CPLANE-10	Field <code>rsvd</code> [12b] legal-space coverage & legality check	FR-CPLANE-10
FEAT-CPLANE-11	Cross-field constraint: <code>start_symbol + num_symbol <= 14</code>	FR-CPLANE-11
FEAT-CPLANE-12	64-bit header pack/unpack round-trip integrity	FR-CPLANE-12
FEAT-CPLANE-13	Reset / power-on default values	FR-CPLANE-13

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-CPLANE	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-CPLANE	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-CPLANE	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-CPLANE	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	17/17	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-002	<code>section_type</code>	enum	one bin per value: {0, 1, 2, 3, 4, 5, 6, 7, 8} (9 bins)	100%
COV-003	<code>section_ext</code>	enum	one bin per value: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11} (11 bins)	100%
COV-005	<code>num_symbol</code>	bucket4	4 range buckets over [1:14] (~3/bucket)	100%
COV-006	<code>num_sections</code>	enum	one bin per value: {1, 2, 4, 8} (4 bins)	100%
COV-008	<code>numerology</code>	enum	one bin per value: {0, 1, 2, 3, 4} (5 bins)	100%
COV-009	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%
COVX-CPLANE-01	<code>cross</code>	cross	legal × illegal region of (<code>start_symbol + num_symbol <= 14</code>)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-CPLANE-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-CPLANE-02	<code>section_type</code> legality: value [0 : 8]	scoreboard + SVA	exercised
CHK-CPLANE-03	<code>section_ext</code> legality: value [1 : 11]	scoreboard + SVA	exercised
CHK-CPLANE-04	<code>start_symbol</code> legality: value [0 : 13]	scoreboard + SVA	exercised
CHK-CPLANE-05	<code>num_symbol</code> legality: value [1 : 14]	scoreboard + SVA	exercised
CHK-CPLANE-06	<code>num_sections</code> legality: value ∈ {1, 2, 4, 8}	scoreboard + SVA	exercised
CHK-CPLANE-07	<code>beam_id</code> legality: value [0 : 65535]	scoreboard + SVA	exercised
CHK-CPLANE-08	<code>numerology</code> legality: value [0 : 4]	scoreboard + SVA	exercised
CHK-CPLANE-09	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-CPLANE-10	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-CPLANE-11	cross invariant: <code>start_symbol + num_symbol <= 14</code>	scoreboard + SVA	exercised
CHK-CPLANE-RT	pack→unpack round-trip == original	scoreboard	exercised

splane — S-Plane PTP/SyncE

Spec: O-RAN.WG4.CUS §9 S-plane IEEE1588v2 PTP + SyncE · fields: 8 · cross-constraints: 0 · coverpoints: 5 · result: **119/119 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-SPLANE-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-SPLANE-01
FEAT-SPLANE-02	Field <code>ptp_msg</code> [4b] legal-space coverage & legality check	FR-SPLANE-02
FEAT-SPLANE-03	Field <code>clock_state</code> [2b] legal-space coverage & legality check	FR-SPLANE-03
FEAT-SPLANE-04	Field <code>sync_e_q1</code> [4b] legal-space coverage & legality check	FR-SPLANE-04
FEAT-SPLANE-05	Field <code>seq</code> [16b] legal-space coverage & legality check	FR-SPLANE-05
FEAT-SPLANE-06	Field <code>timing_err</code> [12b] legal-space coverage & legality check	FR-SPLANE-06
FEAT-SPLANE-07	Field <code>holdover</code> [1b] legal-space coverage & legality check	FR-SPLANE-07
FEAT-SPLANE-08	Field <code>rsvd</code> [21b] legal-space coverage & legality check	FR-SPLANE-08
FEAT-SPLANE-09	64-bit header pack/unpack round-trip integrity	FR-SPLANE-09
FEAT-SPLANE-10	Reset / power-on default values	FR-SPLANE-10

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-SPLANE	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-SPLANE	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-SPLANE	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-DIR-SPLANE	Directed FR / coverage-hole targeting	directed	inline constraint on the hole, removed after closure	targeted bin hit	1/1	PASS
TC-CORN-SPLANE	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	12/12	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-002	<code>ptp_msg</code>	enum	one bin per value: {0, 1, 2, 3, 4, 5} (6 bins)	100%
COV-003	<code>clock_state</code>	enum	one bin per value: {0, 1, 2, 3} (4 bins)	100%
COV-004	<code>sync_e_q1</code>	enum	one bin per value: {0, 2, 4, 11, 15} (5 bins)	100%
COV-005	<code>seq</code>	wrap	min=0, max=65535, mid=[1:65534] (3 bins)	100%
COV-007	<code>holdover</code>	bool	bins {0}, {1} (2 bins)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-SPLANE-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-SPLANE-02	<code>ptp_msg</code> legality: value ∈ {0, 1, 2, 3, 4, 5}	scoreboard + SVA	exercised
CHK-SPLANE-03	<code>clock_state</code> legality: value ∈ {0, 1, 2, 3}	scoreboard + SVA	exercised
CHK-SPLANE-04	<code>sync_e_q1</code> legality: value ∈ {0, 2, 4, 11, 15}	scoreboard + SVA	exercised
CHK-SPLANE-05	<code>seq</code> legality: value [0 : 65535]	scoreboard + SVA	exercised
CHK-SPLANE-06	<code>timing_err</code> legality: value [0 : 3000]	scoreboard + SVA	exercised
CHK-SPLANE-07	<code>holdover</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-SPLANE-08	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-SPLANE-RT	pack→unpack round-trip == original	scoreboard	exercised

mplane — M-Plane NETCONF/YANG

Spec: O-RAN.WG4.MP M-plane NETCONF/YANG O1 · fields: 8 · cross-constraints: 0 · coverpoints: 4 · result: **118/118 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-MPLANE-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-MPLANE-01
FEAT-MPLANE-02	Field <code>netconf_op</code> [3b] legal-space coverage & legality check	FR-MPLANE-02
FEAT-MPLANE-03	Field <code>datastore</code> [2b] legal-space coverage & legality check	FR-MPLANE-03
FEAT-MPLANE-04	Field <code>item_id</code> [16b] legal-space coverage & legality check	FR-MPLANE-04
FEAT-MPLANE-05	Field <code>ant_cal</code> [1b] legal-space coverage & legality check	FR-MPLANE-05
FEAT-MPLANE-06	Field <code>sw_slot</code> [3b] legal-space coverage & legality check	FR-MPLANE-06
FEAT-MPLANE-07	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-MPLANE-07
FEAT-MPLANE-08	Field <code>rsvd</code> [27b] legal-space coverage & legality check	FR-MPLANE-08
FEAT-MPLANE-09	64-bit header pack/unpack round-trip integrity	FR-MPLANE-09
FEAT-MPLANE-10	Reset / power-on default values	FR-MPLANE-10

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-MPLANE	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-MPLANE	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-MPLANE	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-MPLANE	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	12/12	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-002	<code>netconf_op</code>	enum	one bin per value: {0, 1, 2, 3, 4} (5 bins)	100%
COV-003	<code>datastore</code>	enum	one bin per value: {0, 1, 2} (3 bins)	100%
COV-005	<code>ant_cal</code>	bool	bins {0}, {1} (2 bins)	100%
COV-007	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-MPLANE-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-MPLANE-02	<code>netconf_op</code> legality: value ∈ {0, 1, 2, 3, 4}	scoreboard + SVA	exercised
CHK-MPLANE-03	<code>datastore</code> legality: value ∈ {0, 1, 2}	scoreboard + SVA	exercised
CHK-MPLANE-04	<code>item_id</code> legality: value [0 : 65535]	scoreboard + SVA	exercised
CHK-MPLANE-05	<code>ant_cal</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-MPLANE-06	<code>sw_slot</code> legality: value [0 : 7]	scoreboard + SVA	exercised
CHK-MPLANE-07	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-MPLANE-08	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-MPLANE-RT	pack→unpack round-trip == original	scoreboard	exercised

beamforming — Beamforming Engine

Spec: O-RAN.WG4.CUS Section Ext 1/4/5/6 beamforming · fields: 8 · cross-constraints: 1 · coverpoints: 3 · result: **119/119 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-BEAMFORMING-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-BEAMFORMING-01
FEAT-BEAMFORMING-02	Field <code>beam_id</code> [16b] legal-space coverage & legality check	FR-BEAMFORMING-02
FEAT-BEAMFORMING-03	Field <code>num_ports</code> [8b] legal-space coverage & legality check	FR-BEAMFORMING-03
FEAT-BEAMFORMING-04	Field <code>num_layers</code> [4b] legal-space coverage & legality check	FR-BEAMFORMING-04
FEAT-BEAMFORMING-05	Field <code>section_ext</code> [4b] legal-space coverage & legality check	FR-BEAMFORMING-05
FEAT-BEAMFORMING-06	Field <code>codebook_idx</code> [16b] legal-space coverage & legality check	FR-BEAMFORMING-06
FEAT-BEAMFORMING-07	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-BEAMFORMING-07
FEAT-BEAMFORMING-08	Field <code>rsvd</code> [4b] legal-space coverage & legality check	FR-BEAMFORMING-08
FEAT-BEAMFORMING-09	Cross-field constraint: <code>num_layers <= num_ports</code>	FR-BEAMFORMING-09
FEAT-BEAMFORMING-10	64-bit header pack/unpack round-trip integrity	FR-BEAMFORMING-10
FEAT-BEAMFORMING-11	Reset / power-on default values	FR-BEAMFORMING-11

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-BEAMFORMING	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-BEAMFORMING	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-BEAMFORMING	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-BEAMFORMING	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	13/13	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-003	<code>num_ports</code>	enum	one bin per value: {1, 2, 4, 8, 16, 32, 64} (7 bins)	100%
COV-005	<code>section_ext</code>	enum	one bin per value: {1, 4, 5, 6} (4 bins)	100%
COV-007	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%
COVX-BEAMFORMING-01	<code>cross</code>	cross	legal x illegal region of (<code>num_layers <= num_ports</code>)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-BEAMFORMING-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-BEAMFORMING-02	<code>beam_id</code> legality: value [0 : 65535]	scoreboard + SVA	exercised
CHK-BEAMFORMING-03	<code>num_ports</code> legality: value ∈ {1, 2, 4, 8, 16, 32, 64}	scoreboard + SVA	exercised
CHK-BEAMFORMING-04	<code>num_layers</code> legality: value [1 : 8]	scoreboard + SVA	exercised
CHK-BEAMFORMING-05	<code>section_ext</code> legality: value ∈ {1, 4, 5, 6}	scoreboard + SVA	exercised
CHK-BEAMFORMING-06	<code>codebook_idx</code> legality: value [0 : 65535]	scoreboard + SVA	exercised
CHK-BEAMFORMING-07	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-BEAMFORMING-08	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-BEAMFORMING-09	cross invariant: <code>num_layers <= num_ports</code>	scoreboard + SVA	exercised
CHK-BEAMFORMING-RT	pack→unpack round-trip == original	scoreboard	exercised

compression — IQ Compression

Spec: O-RAN.WG4.CUS Annex A IQ compression BFP/mu-law/static · fields: 8 · cross-constraints: 1 · coverpoints: 3 · result: **116/116 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-COMPRESS-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-COMPRESS-01
FEAT-COMPRESS-02	Field <code>method</code> [2b] legal-space coverage & legality check	FR-COMPRESS-02
FEAT-COMPRESS-03	Field <code>iq_width</code> [5b] legal-space coverage & legality check	FR-COMPRESS-03
FEAT-COMPRESS-04	Field <code>comp_width</code> [5b] legal-space coverage & legality check	FR-COMPRESS-04
FEAT-COMPRESS-05	Field <code>exponent</code> [4b] legal-space coverage & legality check	FR-COMPRESS-05
FEAT-COMPRESS-06	Field <code>block_size</code> [8b] legal-space coverage & legality check	FR-COMPRESS-06
FEAT-COMPRESS-07	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-COMPRESS-07
FEAT-COMPRESS-08	Field <code>rsvd</code> [28b] legal-space coverage & legality check	FR-COMPRESS-08
FEAT-COMPRESS-09	Cross-field constraint: <code>comp_width <= iq_width</code>	FR-COMPRESS-09
FEAT-COMPRESS-10	64-bit header pack/unpack round-trip integrity	FR-COMPRESS-10
FEAT-COMPRESS-11	Reset / power-on default values	FR-COMPRESS-11

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-COMPRESS	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-COMPRESS	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-COMPRESS	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-COMPRESS	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	10/10	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-002	<code>method</code>	enum	one bin per value: {0, 1, 2} (3 bins)	100%
COV-004	<code>comp_width</code>	enum	one bin per value: {9, 10, 11, 12, 13, 14, 15, 16} (8 bins)	100%
COV-007	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%
COVX-COMPRESS-01	<code>cross</code>	cross	legal x illegal region of (<code>comp_width <= iq_width</code>)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-COMPRESS-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-COMPRESS-02	<code>method</code> legality: value ∈ {0, 1, 2}	scoreboard + SVA	exercised
CHK-COMPRESS-03	<code>iq_width</code> legality: value == 16	scoreboard + SVA	exercised
CHK-COMPRESS-04	<code>comp_width</code> legality: value ∈ {9, 10, 11, 12, 13, 14, 15, 16}	scoreboard + SVA	exercised
CHK-COMPRESS-05	<code>exponent</code> legality: value [0 : 15]	scoreboard + SVA	exercised
CHK-COMPRESS-06	<code>block_size</code> legality: value [1 : 255]	scoreboard + SVA	exercised
CHK-COMPRESS-07	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-COMPRESS-08	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-COMPRESS-09	cross invariant: <code>comp_width <= iq_width</code>	scoreboard + SVA	exercised
CHK-COMPRESS-RT	pack→unpack round-trip == original	scoreboard	exercised

prach — PRACH Handler

Spec: O-RAN.WG4.CUS Section Type 3 PRACH + 3GPP TS38.211 formats · fields: 9 · cross-constraints: 0 · coverpoints: 4 · result: **118/118 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-PRACH-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-PRACH-01
FEAT-PRACH-02	Field <code>section_type</code> [4b] legal-space coverage & legality check	FR-PRACH-02
FEAT-PRACH-03	Field <code>format</code> [4b] legal-space coverage & legality check	FR-PRACH-03
FEAT-PRACH-04	Field <code>numeroLOGY</code> [4b] legal-space coverage & legality check	FR-PRACH-04
FEAT-PRACH-05	Field <code>fr</code> [1b] legal-space coverage & legality check	FR-PRACH-05
FEAT-PRACH-06	Field <code>zc_root</code> [8b] legal-space coverage & legality check	FR-PRACH-06
FEAT-PRACH-07	Field <code>occasion</code> [6b] legal-space coverage & legality check	FR-PRACH-07
FEAT-PRACH-08	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-PRACH-08
FEAT-PRACH-09	Field <code>rsvd</code> [25b] legal-space coverage & legality check	FR-PRACH-09
FEAT-PRACH-10	64-bit header pack/unpack round-trip integrity	FR-PRACH-10
FEAT-PRACH-11	Reset / power-on default values	FR-PRACH-11

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-PRACH	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-PRACH	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-PRACH	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-PRACH	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	12/12	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-003	<code>format</code>	enum	one bin per value: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9} (10 bins)	100%
COV-004	<code>numeroLOGY</code>	enum	one bin per value: {0, 1, 2, 3, 4} (5 bins)	100%
COV-005	<code>fr</code>	bool	bins {0}, {1} (2 bins)	100%
COV-008	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-PRACH-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-PRACH-02	<code>section_type</code> legality: value == 3	scoreboard + SVA	exercised
CHK-PRACH-03	<code>format</code> legality: value ∈ {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}	scoreboard + SVA	exercised
CHK-PRACH-04	<code>numeroLOGY</code> legality: value [0 : 4]	scoreboard + SVA	exercised
CHK-PRACH-05	<code>fr</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-PRACH-06	<code>zc_root</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-PRACH-07	<code>occasion</code> legality: value [0 : 63]	scoreboard + SVA	exercised
CHK-PRACH-08	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-PRACH-09	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-PRACH-RT	pack→unpack round-trip == original	scoreboard	exercised

mimo_massive — Massive MIMO

Spec: O-RAN.WG4.CUS Sect Ext 5/6 + 3GPP massive MIMO 64T64R · fields: 8 · cross-constraints: 2 · coverpoints: 3 · result: **120/120 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-MIMO_MASSIVE-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-MIMO_MASSIVE-01
FEAT-MIMO_MASSIVE-02	Field <code>ant_cfg</code> [8b] legal-space coverage & legality check	FR-MIMO_MASSIVE-02
FEAT-MIMO_MASSIVE-03	Field <code>num_layers</code> [4b] legal-space coverage & legality check	FR-MIMO_MASSIVE-03
FEAT-MIMO_MASSIVE-04	Field <code>tdd_cfg</code> [3b] legal-space coverage & legality check	FR-MIMO_MASSIVE-04
FEAT-MIMO_MASSIVE-05	Field <code>rank</code> [4b] legal-space coverage & legality check	FR-MIMO_MASSIVE-05
FEAT-MIMO_MASSIVE-06	Field <code>precoder_idx</code> [8b] legal-space coverage & legality check	FR-MIMO_MASSIVE-06
FEAT-MIMO_MASSIVE-07	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-MIMO_MASSIVE-07
FEAT-MIMO_MASSIVE-08	Field <code>rsvd</code> [25b] legal-space coverage & legality check	FR-MIMO_MASSIVE-08
FEAT-MIMO_MASSIVE-09	Cross-field constraint: <code>num_layers</code> <= <code>ant_cfg</code>	FR-MIMO_MASSIVE-09
FEAT-MIMO_MASSIVE-10	Cross-field constraint: <code>rank</code> <= <code>num_layers</code>	FR-MIMO_MASSIVE-10
FEAT-MIMO_MASSIVE-11	64-bit header pack/unpack round-trip integrity	FR-MIMO_MASSIVE-11
FEAT-MIMO_MASSIVE-12	Reset / power-on default values	FR-MIMO_MASSIVE-12

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-MIMO_MASSIVE	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-MIMO_MASSIVE	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-MIMO_MASSIVE	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-MIMO_MASSIVE	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	14/14	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-002	<code>ant_cfg</code>	enum	one bin per value: {1, 2, 4, 8, 16, 32, 64} (7 bins)	100%
COV-004	<code>tdd_cfg</code>	enum	one bin per value: {0, 1, 2, 3, 4, 5, 6} (7 bins)	100%
COV-007	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%
COVX-MIMO_MASSIVE-01	<code>cross</code>	cross	legal × illegal region of (<code>num_layers</code> <= <code>ant_cfg</code>)	100%
COVX-MIMO_MASSIVE-02	<code>cross</code>	cross	legal × illegal region of (<code>rank</code> <= <code>num_layers</code>)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-MIMO_MASSIVE-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-02	<code>ant_cfg</code> legality: value ∈ {1, 2, 4, 8, 16, 32, 64}	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-03	<code>num_layers</code> legality: value [1 : 8]	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-04	<code>tdd_cfg</code> legality: value [0 : 6]	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-05	<code>rank</code> legality: value [1 : 8]	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-06	<code>precoder_idx</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-07	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-08	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-09	cross invariant: <code>num_layers</code> <= <code>ant_cfg</code>	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-10	cross invariant: <code>rank</code> <= <code>num_layers</code>	scoreboard + SVA	exercised
CHK-MIMO_MASSIVE-RT	pack→unpack round-trip == original	scoreboard	exercised

bwp — Bandwidth-Part Manager

Spec: O-RAN.WG4.CUS BWP + 3GPP TS38.211 FR1/FR2 numerology · fields: 8 · cross-constraints: 0 · coverpoints: 6 · result: **118/118 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-BWP-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-BWP-01
FEAT-BWP-02	Field <code>bwp_id</code> [2b] legal-space coverage & legality check	FR-BWP-02
FEAT-BWP-03	Field <code>numerology</code> [4b] legal-space coverage & legality check	FR-BWP-03
FEAT-BWP-04	Field <code>bw_mhz</code> [9b] legal-space coverage & legality check	FR-BWP-04
FEAT-BWP-05	Field <code>fr</code> [1b] legal-space coverage & legality check	FR-BWP-05
FEAT-BWP-06	Field <code>active</code> [1b] legal-space coverage & legality check	FR-BWP-06
FEAT-BWP-07	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-BWP-07
FEAT-BWP-08	Field <code>rsvd</code> [35b] legal-space coverage & legality check	FR-BWP-08
FEAT-BWP-09	64-bit header pack/unpack round-trip integrity	FR-BWP-09
FEAT-BWP-10	Reset / power-on default values	FR-BWP-10

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-BWP	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-BWP	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-BWP	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-BWP	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	12/12	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-002	<code>bwp_id</code>	enum	one bin per value: {0, 1, 2, 3} (4 bins)	100%
COV-003	<code>numerology</code>	enum	one bin per value: {0, 1, 2, 3, 4} (5 bins)	100%
COV-004	<code>bw_mhz</code>	enum	one bin per value: {5, 10, 15, 20, 25, 40, 50, 60, 80, 100, 200, 400} (12 bins)	100%
COV-005	<code>fr</code>	bool	bins {0}, {1} (2 bins)	100%
COV-006	<code>active</code>	bool	bins {0}, {1} (2 bins)	100%
COV-007	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-BWP-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-BWP-02	<code>bwp_id</code> legality: value [0 : 3]	scoreboard + SVA	exercised
CHK-BWP-03	<code>numerology</code> legality: value [0 : 4]	scoreboard + SVA	exercised
CHK-BWP-04	<code>bw_mhz</code> legality: value ∈ {5, 10, 15, 20, 25, 40, 50, 60, 80, 100, 200, 400}	scoreboard + SVA	exercised
CHK-BWP-05	<code>fr</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-BWP-06	<code>active</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-BWP-07	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-BWP-08	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-BWP-RT	pack→unpack round-trip == original	scoreboard	exercised

mmwave — mmWave FR2 Handler

Spec: 3GPP TS38.211 FR2 mmWave n257/n258/n260/n261, mu=2/3/4 · fields: 8 · cross-constraints: 0 · coverpoints: 5 · result: **118/118 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-MMWAVE-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-MMWAVE-01
FEAT-MMWAVE-02	Field <code>band</code> [3b] legal-space coverage & legality check	FR-MMWAVE-02
FEAT-MMWAVE-03	Field <code>numerology</code> [4b] legal-space coverage & legality check	FR-MMWAVE-03
FEAT-MMWAVE-04	Field <code>scs</code> [2b] legal-space coverage & legality check	FR-MMWAVE-04
FEAT-MMWAVE-05	Field <code>ssb_period</code> [4b] legal-space coverage & legality check	FR-MMWAVE-05
FEAT-MMWAVE-06	Field <code>beam_id</code> [16b] legal-space coverage & legality check	FR-MMWAVE-06
FEAT-MMWAVE-07	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-MMWAVE-07
FEAT-MMWAVE-08	Field <code>rsvd</code> [23b] legal-space coverage & legality check	FR-MMWAVE-08
FEAT-MMWAVE-09	64-bit header pack/unpack round-trip integrity	FR-MMWAVE-09
FEAT-MMWAVE-10	Reset / power-on default values	FR-MMWAVE-10

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-MMWAVE	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-MMWAVE	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-MMWAVE	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-MMWAVE	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	12/12	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-002	<code>band</code>	enum	one bin per value: {0, 1, 2, 3} (4 bins)	100%
COV-003	<code>numerology</code>	enum	one bin per value: {2, 3, 4} (3 bins)	100%
COV-004	<code>scs</code>	enum	one bin per value: {0, 1, 2} (3 bins)	100%
COV-005	<code>ssb_period</code>	enum	one bin per value: {0, 1, 2, 3, 4, 5} (6 bins)	100%
COV-007	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-MMWAVE-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-MMWAVE-02	<code>band</code> legality: value ∈ {0, 1, 2, 3}	scoreboard + SVA	exercised
CHK-MMWAVE-03	<code>numerology</code> legality: value [2 : 4]	scoreboard + SVA	exercised
CHK-MMWAVE-04	<code>scs</code> legality: value ∈ {0, 1, 2}	scoreboard + SVA	exercised
CHK-MMWAVE-05	<code>ssb_period</code> legality: value ∈ {0, 1, 2, 3, 4, 5}	scoreboard + SVA	exercised
CHK-MMWAVE-06	<code>beam_id</code> legality: value [0 : 65535]	scoreboard + SVA	exercised
CHK-MMWAVE-07	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-MMWAVE-08	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-MMWAVE-RT	pack→unpack round-trip == original	scoreboard	exercised

laa — LAA Handler

Spec: O-RAN.WG4.CUS Section Type 5 LAA + LBT Cat1-4 · fields: 9 · cross-constraints: 0 · coverpoints: 5 · result: **116/116 PASS** · func cov 100% · code stmt 100% · branch(eff) 100%

FEATURES & REQUIREMENTS

Feature	Description	Requirement
FEAT-LAA-01	Field <code>version</code> [4b] legal-space coverage & legality check	FR-LAA-01
FEAT-LAA-02	Field <code>section_type</code> [4b] legal-space coverage & legality check	FR-LAA-02
FEAT-LAA-03	Field <code>lbt_cat</code> [3b] legal-space coverage & legality check	FR-LAA-03
FEAT-LAA-04	Field <code>lbt_result</code> [1b] legal-space coverage & legality check	FR-LAA-04
FEAT-LAA-05	Field <code>burst_type</code> [1b] legal-space coverage & legality check	FR-LAA-05
FEAT-LAA-06	Field <code>cap</code> [2b] legal-space coverage & legality check	FR-LAA-06
FEAT-LAA-07	Field <code>section_ext</code> [4b] legal-space coverage & legality check	FR-LAA-07
FEAT-LAA-08	Field <code>seq</code> [8b] legal-space coverage & legality check	FR-LAA-08
FEAT-LAA-09	Field <code>rsvd</code> [37b] legal-space coverage & legality check	FR-LAA-09
FEAT-LAA-10	64-bit header pack/unpack round-trip integrity	FR-LAA-10
FEAT-LAA-11	Reset / power-on default values	FR-LAA-11

TEST SCENARIOS (EXECUTED TEST CLASSES)

Test case	Intent	Class	Stimulus (constrained-random)	Pass criteria	Runs	Status
TC-SMOKE-LAA	Directed sanity, fixed seeds 1/2/3	smoke	constrained-random (Z3 model), seed-locked	legal txn, UVM_ERROR=0, TEST PASSED	3/3	PASS
TC-RAND-LAA	Constrained-random soak (100 seeds)	random	randomize() with {} legal-space + cross constraints	0 legality/cross violations across all seeds	100/100	PASS
TC-NEG-LAA	Error injection (illegal fields)	negative	inject_illegal — constrained illegal space	violation DETECTED = EXPECTED_FAILURE_DETECTED	3/3	PASS
TC-CORN-LAA	Corner / boundary (min/max/wrap)	corner	boundary-weighted constraints (dist{})	all boundary bins hit, legal	10/10	PASS

FUNCTIONAL COVERAGE POINTS

Cov ID	Field	Kind	Bins	Closure
COV-003	<code>lbt_cat</code>	enum	one bin per value: {1, 2, 3, 4} (4 bins)	100%
COV-004	<code>lbt_result</code>	bool	bins {0}, {1} (2 bins)	100%
COV-005	<code>burst_type</code>	bool	bins {0}, {1} (2 bins)	100%
COV-006	<code>cap</code>	enum	one bin per value: {0, 1, 2, 3} (4 bins)	100%
COV-008	<code>seq</code>	wrap	min=0, max=255, mid=[1:254] (3 bins)	100%

CHECKERS & ASSERTIONS

Check ID	Invariant	Mechanism	Status
CHK-LAA-01	<code>version</code> legality: value == 1	scoreboard + SVA	exercised
CHK-LAA-02	<code>section_type</code> legality: value == 5	scoreboard + SVA	exercised
CHK-LAA-03	<code>lbt_cat</code> legality: value ∈ {1, 2, 3, 4}	scoreboard + SVA	exercised
CHK-LAA-04	<code>lbt_result</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-LAA-05	<code>burst_type</code> legality: value [0 : 1]	scoreboard + SVA	exercised
CHK-LAA-06	<code>cap</code> legality: value [0 : 3]	scoreboard + SVA	exercised
CHK-LAA-07	<code>section_ext</code> legality: value == 3	scoreboard + SVA	exercised
CHK-LAA-08	<code>seq</code> legality: value [0 : 255]	scoreboard + SVA	exercised
CHK-LAA-09	<code>rsvd</code> legality: value == 0	scoreboard + SVA	exercised
CHK-LAA-RT	pack→unpack round-trip == original	scoreboard	exercised

RAL — uvm_reg Register Model (M-Plane RU config)

Native UVM uvm_reg · 5 registers · front-door via uvm_reg_adapter + auto-predict · result: **PASS** · 163 ops · 0 err (Verilator 5.050)

REGISTER TEST SCENARIOS

Test case	Intent	Access	Pass criteria	Status
TC-RAL-RST	Reset-value mirror check (all regs)	—	mirror == reset value (e.g. comp.width==16)	PASS
TC-RAL-RO	RO version read-back	RO	reads 0x0A02 (major 0x02 / minor 0x0A)	PASS
TC-RAL-RW	20× per-reg randomize→update→read→mirror-compare	RW	read data == desired mirror, 0 mismatch	PASS
TC-RAL-WP	RO write-protect	RO	write to version leaves value unchanged	PASS

REGISTER MAP

Register	Offset	Access	Fields
ru_ctrl_reg	0x00	RW	enable[0], mode[2:1], antmap[10:3]
ru_version_reg	0x04	RO	major[7:0]=0x02, minor[15:8]=0x0A
ru_numant_reg	0x08	RW	n_tx[6:0], n_rx[14:8]
ru_comp_reg	0x0C	RW	method[1:0], width[6:2]=16
ru_bwp_reg	0x10	RW	bwp_id[1:0], numerology[5:2], active[6]

4 · Coverage report — summary

FUNCTIONAL & CODE COVERAGE (GATE 8), PER COMPONENT

Component	tests	func cov	func holes	code stmt	branch raw	waived	branch eff	toggle	GATE 8
ecpri_transport	40/40	100%	0	100%	75%	6	100%	54.7%	PASS_WITH_WAIVERS
cpri_eth	122/122	100%	0	100%	75%	6	100%	43.8%	PASS_WITH_WAIVERS
uplane	123/123	100%	0	100%	75%	6	100%	60.9%	PASS_WITH_WAIVERS
cplane	123/123	100%	0	100%	75%	6	100%	57.8%	PASS_WITH_WAIVERS
splane	119/119	100%	0	100%	75%	6	100%	57.8%	PASS_WITH_WAIVERS
mplane	118/118	100%	0	100%	75%	6	100%	50.0%	PASS_WITH_WAIVERS
beamforming	119/119	100%	0	100%	75%	6	100%	45.3%	PASS_WITH_WAIVERS
compression	116/116	100%	0	100%	75%	6	100%	35.9%	PASS_WITH_WAIVERS
prach	118/118	100%	0	100%	75%	6	100%	46.9%	PASS_WITH_WAIVERS
mimo_massive	120/120	100%	0	100%	75%	6	100%	32.8%	PASS_WITH_WAIVERS
bwp	118/118	100%	0	100%	75%	6	100%	35.9%	PASS_WITH_WAIVERS
mmwave	118/118	100%	0	100%	75%	6	100%	50.0%	PASS_WITH_WAIVERS
laa	116/116	100%	0	100%	75%	6	100%	25.0%	PASS_WITH_WAIVERS

Functional coverage = merged COV-### bins (per-field + cross) across smoke+random+directed+corner runs, 100% with 0 open holes. Code: statement 100%; branch 100% effective after waiving TB-defensive branches (0 DUT branches uncovered). Toggle reported for completeness (not a GATE-8 criterion). Native covergroup get_coverage() not computed by Verilator 5.050 — real covergroup % on VCS/Questa/Xcelium; protocol functional coverage is 100% on identical bins.

Evidence: components/<slug>/spec.json regression/suite_regression_full.json regression/code_cov.json regression/combined_regression.json lane2/ral/ral_result.json . Companion: FINAL_REGRESSION_COVERAGE_REPORT.html, COVERAGE_DASHBOARD.html, VERIFICATION_REFERENCE, FUNCTIONAL_DESIGN. Open (non-blocking): GATE-0 requirements DERIVED (pin O-RAN.WG4.CUS R004-v16). · AVIK VIP FACTORY — O-RAN VIP Suite.